

Week 09: When fine-tuning helps

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Choose among a clearer prompt, current evidence and model adaptation.

Context and teaching plan

A college assistant uses the correct format but misses a revised policy.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Prompting

Change instructions or examples for the task without changing weights.

Teacher question: Where does prompting appear in the weekly case, and what could go wrong if we omit it?

RAG

Supply up-to-date evidence at inference time; useful for changing facts and traceable answers.

Teacher question: Where does rag appear in the weekly case, and what could go wrong if we omit it?

Fine-tuning

Train on examples to change learned behaviour; separate development from final test cases.

Teacher question: Where does fine-tuning appear in the weekly case, and what could go wrong if we omit it?

LoRA

Update a small trainable low-rank component instead of all model weights.

Teacher question: Where does lora appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: Base 6/10; prompt 8/10; tuned 9/10 on practice cases.
- Change or question: On fresh cases tuned score is 6/10.
- Supported observation: The practice gain alone does not prove better generalization.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The revised policy problem: Practice results say a tuned model wins 9/10, but a revised policy arrived yesterday and final cases show 6/10. What should change first?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Prompting: analogy, mechanism and check

Familiar picture: A clearer recipe can change a cook's output without retraining the cook.

Familiar part	Technical meaning
Recipe	prompt
Example plate	few-shot example
Taste check	task evaluation

Where this analogy breaks: Instructions can fail; a prompt does not update facts or guarantee conformance.

Actual idea: Change instructions or examples for the task without changing weights.

Work it aloud: On ten practice cases, a clarified prompt improves 6/10 to 8/10; record the cases.

Ask before revealing: When is this an inexpensive first trial?

Supported response: When the failure concerns task instructions or format.

Look for this misconception: A prompt can permanently teach yesterday's revised policy.

RAG: analogy, mechanism and check

Familiar picture: A clerk opens the current rulebook instead of relying on memory.

Familiar part	Technical meaning
Rulebook	updated policy
Page search	retrieval

Quoted line	grounded answer
Where this analogy breaks:	Retrieved text still needs access, relevance and claim checks.
Actual idea:	Supply up-to-date evidence at inference time; useful for changing facts and traceable answers.
Work it aloud:	Replace P3 with its approved revised version, reindex and retrieve it for the question.
Ask before revealing:	Where should a changed policy be updated first?
Supported response:	In the authoritative document and retrieval index.
Look for this misconception:	Retrieval changes model weights.

Fine-tuning: analogy, mechanism and check

Familiar picture: A musician practises many examples to change a habit.

Familiar part	Technical meaning
Practice pieces	labelled data
Habit change	weight update
New concert	final test

Where this analogy breaks: Small practice gains can reflect overfitting; tuning is not a live fact store.

Actual idea: Train on examples to change learned behaviour; separate development from final test cases.

Work it aloud: Base 6/10, prompted 8/10, tuned 9/10 in practice; final tuned 6/10. Investigate.

Ask before revealing: Does the practice result prove tuning should ship?

Supported response: No; the fresh test and failure analysis matter.

Look for this misconception: Fine-tuning is guaranteed to beat a good prompt.

LoRA: analogy, mechanism and check

Familiar picture: A small adapter changes a machine's behaviour without replacing its engine.

Familiar part	Technical meaning
Engine	frozen base weights
Adapter	low-rank matrices
New behaviour	tuned output

Where this analogy breaks: Low trainable parameter count does not remove data, compute or evaluation needs.

Actual idea: Update a small trainable low-rank component instead of all model weights.

Work it aloud: LoRA trains small matrices while freezing most base weights; no live tuning is required in this lab.

Ask before revealing: What does LoRA reduce?

Supported response: The trainable parameter count compared with full fine-tuning.

Look for this misconception: LoRA retrieves current policy documents automatically.

New situation for discussion

The system returns the right fact but the wrong JSON field names. Which adaptation is the simplest first trial?

Memory cue: Diagnose the failure, choose prompt or RAG or tuning, and test fresh cases.

Guided laboratory

Analyze a supplied results pack without training a model.

1. Compare provided base, prompted and adapted predictions
2. Score each on development cases
3. Inspect a held-out final set once
4. Recommend a method for a policy update

Individual change: Change one threshold and note why the final set must remain untouched. Expected evidence: Comparison table with recommendation and uncertainty.

Checks and common error

Measure: Accuracy with numerator and denominator on separate cases. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: Fine-tuning is the right way to add a policy changed yesterday.

Correction: Update the source document and retrieval index; weight adaptation is poor for rapidly changing facts.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?

- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Hu et al., LoRA. <https://arxiv.org/abs/2106.09685>

NVIDIA, NeMo framework overview.

<https://docs.nvidia.com/nemo-framework/user-guide/latest/overview.html>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Finetune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO3, LO4 | 2 hours concepts + 2 hours guided practice

The idea in plain language

Choose a method based on the failure. A clearer prompt can improve instructions. RAG supplies current facts. Fine-tuning changes learned behavior from examples. LoRA trains low-rank update matrices while freezing most base parameters. Small adaptation datasets still need separate evaluation.

A worked example

The supplied practice scores are base 6/10, prompted 8/10, tuned 9/10. On final cases prompted is 7/10 and tuned is 6/10. These small samples do not prove a general advantage for tuning. A changed policy should first be updated in the evidence source.

Practice before reading the answers

1. Which method suits a policy that changed yesterday? (4 suggested practice marks)
2. What does LoRA reduce compared with full fine-tuning? (4 suggested practice marks)
3. Can 9/10 on practice cases prove deployment quality? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. Update the authoritative document and retrieval data. Fine-tuning is not a dependable store for rapidly changing facts.

2. The number of trainable parameters by using low-rank updates. It does not remove the need for data quality or evaluation.

3. No. Use a separate test set, examine failures and report the small denominator.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

O1 OpenAI: Evaluation best practices

<https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Study focus: Task-specific testing, repeated outputs and separate evaluation cases.

N2 NVIDIA: NeMo Framework overview

<https://docs.nvidia.com/nemo-framework/user-guide/latest/overview.html>

Study focus: Instructor background on model customization; no NVIDIA setup required in core labs.

N3 NVIDIA: Supported PEFT methods: LoRA

https://docs.nvidia.com/nemo-framework/user-guide/25.02/sft_peft/supported_methods.html

Study focus: Frozen base weights and low-rank updates. Read the concept section; live setup is optional.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.